AI.103

Machine Learning

COURSE OVERVIEW

Foundations of machine learning: supervised, unsupervised, and reinforcement learning; data, models, training, and evaluation.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

ORIENTATION: MACHINES THAT LEARN

KEY TOPICS

- AI vs. ML vs. deep learning
- The learning problem: data → model
- Training, validation, test splits
- Types of ML: supervised, unsupervised, RL

TEACHING FOCUS

Define machine learning and its place in AI.

KEY TAKEAWAY Learning is improving performance through experience

2

SESSION

60 MINUTES

SUPERVISED LEARNING — REGRESSION

KEY TOPICS

- Linear regression and least squares
- Polynomial features and overfitting
- Gradient descent optimization
- Evaluating regression models

TEACHING FOCUS

Fit continuous-output models to data.

KEY TAKEAWAY Predict numbers from features.

3

SESSION

60 MINUTES

SUPERVISED LEARNING — CLASSIFICATION

KEY TOPICS

- Logistic regression
- Decision trees and random forests
- k-Nearest neighbors
- Support vector machines

TEACHING FOCUS

Assign inputs to discrete classes.

KEY TAKEAWAY Draw boundaries that separate classes.

4

SESSION

60 MINUTES

UNSUPERVISED LEARNING

KEY TOPICS

- k-Means clustering
- Hierarchical clustering
- Principal Component Analysis (PCA)
- Dimensionality reduction and visualization

TEACHING FOCUS

Discover hidden structure without labels.

KEY TAKEAWAY Find patterns where labels do not exist.

5

SESSION

60 MINUTES

MODEL EVALUATION & SELECTION

KEY TOPICS

- Accuracy, precision, recall, F1
- Confusion matrix
- Cross-validation
- Bias–variance tradeoff

TEACHING FOCUS

Choose models that generalize, not memorize.

KEY TAKEAWAY Generalization is the true test of learning.

6

SESSION

60 MINUTES

ML IN ROBOTICS & ETHICS

KEY TOPICS

- ML for perception and control
- Data bias and fairness
- Course recap and assessment
- Preparation for neural networks

TEACHING FOCUS

Apply ML responsibly to robotic systems.

KEY TAKEAWAY Data-driven robots must be fair and robust.